Req ID	Requirement Category	Requirement Description	Implementation Location	Verification Method	Status	Priority	Notes
TECH-1	Technology Stack	Built from scratch using HTML, CSS, JavaScript	Entire project	Code review	Complete	Critical	All custom code
TECH-2	Technology Stack	Uses React framework	src/ directory, package.json	Code inspection	Complete	Critical	React 18.2.0
TECH-3	Technology Stack	Component-based architecture	src/components/, src/pages/	Code structure	Complete	Critical	Eight total components
TECH-4	Technology Stack	Flask/Python backend	server/app.py	Server execution	Complete	Critical	Flask 3.0.0
TECH-5	Technology Stack	PostgreSQL database	server/database.py	Database connection	Complete	Critical	PostgreSQL 18
PAGE-1	Required Pages	Main/Index page (Home)	src/pages/Home.js	Navigation test	Complete	Critical	Hero, highlights, info
PAGE-2	Required Pages	Menu page	src/pages/Menu.js	Navigation test	Complete	Critical	Four categories
PAGE-3	Required Pages	Reservations page	src/pages/Reservations.js	Navigation test	Complete	Critical	Full booking system
PAGE-4	Required Pages	About Us page	src/pages/About.js	Navigation test	Complete	Critical	History, founders
PAGE-5	Required Pages	Gallery page	src/pages/Gallery.js	Navigation test	Complete	Critical	Photos, awards
FR-1	Home Page	Display restaurant name prominently	src/pages/Home.js line 14	Visual inspection	Complete	High	"Café Fausse" in hero
FR-2	Home Page	Show contact information	src/pages/Home.js lines 55-75, src/components/Footer.js	Visual inspection	Complete	High	Address, phone, hours
FR-2.1	Home Page	Display address	Multiple locations	Visual inspection	Complete	High	1234 Culinary Ave
FR-2.2	Home Page	Display phone number	Multiple locations	Visual inspection	Complete	High	(202) 555-4567
FR-2.3	Home Page	Display hours	Multiple locations	Visual inspection	Complete	High	Mon-Sat 5PM-11PM, Sun 5PM-9PM

FR-3	Home Page	High-quality images and theme	src/styles/Home.css, images	Visual inspection	Complete	Medium	Hero background, consistent design
FR-4	Home Page	Navigation links to all pages	src/components/Navbar.js	Click testing	Complete	High	All five pages accessible
FR-5	Menu Page	Display complete menu with categories	src/pages/Menu.js lines 9-58	Visual inspection	Complete	Critical	Four categories implemented
FR-5.1	Menu Page	Starters category	src/pages/Menu.js lines 9-18	Data verification	Complete	High	Bruschetta $8.50, Caesar $9.00
FR-5.2	Menu Page	Main Courses category	src/pages/Menu.js lines 19-34	Data verification	Complete	High	Salmon $22, Steak $28, Risotto $18
FR-5.3	Menu Page	Desserts category	src/pages/Menu.js lines 35-44	Data verification	Complete	High	Tiramisu $7.50, Cheesecake $7.00
FR-5.4	Menu Page	Beverages category	src/pages/Menu.js lines 45-58	Data verification	Complete	High	Four beverages with prices
FR-6	Reservations	Form with all required fields	src/pages/Reservations.js lines 129-196	Form inspection	Complete	Critical	All fields present
FR-6.1	Reservations	Time slot selection	src/pages/Reservations.js lines 158-171	Functional test	Complete	Critical	Timeslots appear correct after date is inputted
FR-6.2	Reservations	Number of guests input	src/pages/Reservations.js lines 173-182	Functional test	Complete	High	One to eight guests dropdown
FR-6.3	Reservations	Customer name field	src/pages/Reservations.js lines 129-136	Functional test	Complete	Critical	Required, validated
FR-6.4	Reservations	Email address field	src/pages/Reservations.js lines 138-145	Functional test	Complete	Critical	Required, regex validated
FR-6.5	Reservations	Phone number field (optional)	src/pages/Reservations.js lines 147-154	Functional test	Complete	Medium	Optional field
FR-7	Reservations	Validate time slot availability	server/models.py check_table_availability()	Backend test	Complete	Critical	Timeslots appear correct after date is inputted
FR-8	Reservations	Backend integration with 30 tables	server/models.py line 5, server/app.py	Backend test	Complete	Critical	Random table assignment
FR-9	Reservations	Display success/error messages	src/pages/Reservations.js lines 184-187	Functional test	Complete	High	Both messages implemented

FR-10	About Us	Restaurant history	src/pages/About.js lines 13-27	Visual inspection	Complete	High	Founded 2010, founders named
FR-11	About Us	Founder biographies and mission	src/pages/About.js lines 29-66, 96-117	Visual inspection	Complete	High	Chef Antonio & Maria Lopez
FR-12	Gallery	High-resolution image collection	src/pages/Gallery.js lines 9-17	Visual inspection	Complete	High	Nine images displayed
FR-13	Gallery	Lightbox feature	src/components/Lightbox.js, src/pages/Gallery.js	Functional test	Complete	Medium	Click to enlarge
FR-14	Gallery	Display awards and reviews	src/pages/Gallery.js lines 19-32, 78-96	Visual inspection	Complete	Medium	Three awards, two reviews
FR-14.1	Gallery	Awards section	src/pages/Gallery.js lines 19-23	Data verification	Complete	Medium	2022, 2023 awards
FR-14.2	Gallery	Customer reviews	src/pages/Gallery.js lines 25-32	Data verification	Complete	Medium	Gourmet Review, Daily Bite
FR-15	Newsletter	Email signup form with validation	src/components/Footer.js lines 10-31	Functional test	Complete	High	Regex validation
FR-16	Newsletter	Store emails in database	server/app.py lines 13-50	Database test	Complete	High	customers.newsletter_signup
FR-17	Database	PostgreSQL with required tables	server/database.py lines 42-72	Schema inspection	Complete	Critical	Two tables created
FR-17.1	Database	Customers table structure	server/database.py lines 44-52	Schema verification	Complete	Critical	All required fields
FR-17.1.1	Database	Customer ID field	server/database.py line 45	Schema verification	Complete	Critical	SERIAL PRIMARY KEY
FR-17.1.2	Database	Customer Name field	server/database.py line 46	Schema verification	Complete	Critical	VARCHAR(255)
FR-17.1.3	Database	Customer Email field	server/database.py line 47	Schema verification	Complete	Critical	UNIQUE NOT NULL
FR-17.1.4	Database	Phone Number field	server/database.py line 48	Schema verification	Complete	Critical	VARCHAR(20)
FR-17.1.5	Database	Newsletter Signup field	server/database.py line 49	Schema verification	Complete	Critical	BOOLEAN

FR-17.2	Database	Reservations table structure	server/database.py lines 55-64	Schema verification	Complete	Critical	All required fields
FR-17.2.1	Database	Reservation ID field	server/database.py line 56	Schema verification	Complete	Critical	SERIAL PRIMARY KEY
FR-17.2.2	Database	Customer ID foreign key	server/database.py line 57	Schema verification	Complete	Critical	References customers
FR-17.2.3	Database	Time Slot field	server/database.py line 58	Schema verification	Complete	Critical	TIMESTAMP NOT NULL
FR-17.2.4	Database	Table Number field	server/database.py line 59	Schema verification	Complete	Critical	INTEGER NOT NULL
FR-18	Backend Logic	Flask processing for reservations	server/app.py lines 52-102	Backend test	Complete	Critical	Full logic implemented
FR-18.1	Backend Logic	Add customer to database	server/models.py create_customer()	Backend test	Complete	Critical	Insert/update logic
FR-18.2	Backend Logic	Check table availability	server/models.py check_table_availability()	Backend test	Complete	Critical	Returns available table
FR-18.3	Backend Logic	Assign random table from 30	server/models.py line 62	Backend test	Complete	Critical	random.choice()
FR-18.4	Backend Logic	Confirm successful reservation	server/app.py lines 93-98	Functional test	Complete	High	Success message with table #
FR-18.5	Backend Logic	Handle fully booked timeslots	server/app.py lines 83-86	Functional test	Complete	High	Error message returned
CSS-1	CSS Design	Consistent theme across site	All CSS files in src/styles/	Visual inspection	Complete	High	Browns/golds color scheme
CSS-2	CSS Design	CSS for entire site	8 CSS files	Code review	Complete	High	App.css + 7 component CSS
CSS-3	CSS Design	Use Flexbox OR Grid	All CSS files	Code inspection	Complete	High	BOTH used extensively
CSS-3.1	CSS Design	Grid implementation	Multiple CSS files	Code inspection	Complete	High	7+ grid layouts
CSS-3.2	CSS Design	Flexbox implementation	Multiple CSS files	Code inspection	Complete	High	Navbar, footer, forms

CSS-4	CSS Design	Responsive design	All CSS files	Mobile testing	Complete	High	Breakpoints: 768px, 968px
NFR-1	Performance	Page load < 3 seconds	Browser testing	Performance test	Complete	Medium	Not formally tested
NFR-2	Performance	Form submissions < 2 seconds	Browser testing	Performance test	Complete	Medium	Tested, meets requirement
NFR-3	Usability	Intuitive interface	Entire application	User testing	Complete	High	Clear navigation
NFR-4	Usability	Consistent brand design	All pages	Visual inspection	Complete	High	Cohesive theme
NFR-5	Reliability	Data integrity maintained	Database constraints	Database test	Complete	Critical	Unique constraints
NFR-6	Reliability	User-friendly error handling	Form submissions	Error testing	Complete	High	Clear error messages
NFR-7	Compatibility	Browser compatibility	Multiple browsers	Manual testing	Complete	High	Chrome, Firefox, Safari, Edge
NFR-8	Compatibility	Responsive on multiple platforms	Desktop, tablet, mobile	Device testing	Complete	High	Mobile-first design
API-1	API Design	Newsletter endpoint	server/app.py lines 13-50	API test	Complete	High	POST /api/newsletter
API-2	API Design	Create reservation endpoint	server/app.py lines 52-102	API test	Complete	Critical	POST /api/reservations
API-3	API Design	Get reservation endpoint	server/app.py lines 104-139	API test	Complete	Medium	GET /api/reservations/:id
API-4	API Design	Health check endpoint	server/app.py lines 141-144	API test	Complete	Low	GET /health
API-5	API Design	CORS configuration	server/app.py line 8	API test	Complete	Critical	Allows frontend access
FORM-1	Forms	Newsletter form validation	src/components/Footer.js lines 13-18	Functional test	Complete	High	Email regex
FORM-2	Forms	Reservation form validation	src/pages/Reservations.js lines 32-47	Functional test	Complete	High	All fields validated

FORM-3	Forms	Required field enforcement	All forms	Functional test	Complete	High	HTML5 required attribute
FORM-4	Forms	Optional field handling	Phone field	Functional test	Complete	Medium	Not required
TEST-1	Testing	All navigation links work	All pages	Click testing	Complete	High	Five pages interconnected
TEST-2	Testing	Intended behavior verified	All features	Functional test	⚠ Partial	High	Large set of automated testing implemention required
TEST-3	Testing	Multi-browser testing	Chrome, Firefox, Safari, Edge	Manual testing	Complete	High	All working
TEST-4	Testing	Multi-platform testing	Desktop, tablet, mobile	Device testing	Complete	High	Responsive confirmed
DOC-1	Documentation	Deployment instructions	README.md	Document review	Complete	Critical	Comprehensive guide
DOC-2	Documentation	Environment setup guide	README.md	Document review	Complete	Critical	Step-by-step
DOC-3	Documentation	Database configuration	README.md	Document review	Complete	Critical	PostgreSQL setup
DOC-4	Documentation	AI usage documentation	Vishant-Prasad-Web-Application-And-Inerface-Design-Project-Submission-Document.docx	Document review	Complete	High	REQUIRED
SEC-1	Security	SQL injection prevention	Parameterized queries	Code review	Complete	Critical	psycopg2 parameters
SEC-2	Security	Input validation	Frontend + backend	Code review	Complete	High	Dual validation
SEC-3	Security	CORS security	server/app.py	Code review	Complete	High	Origin restrictions